

SCE-009 — FROM WHY TO AUTONOMOUS AI

Why Meaningful Autonomous Intelligence May Require Internal Question Emergence

Structural Cognitive Emergence (SCE)

Sizhe Tan
with chatGPT (OpenAI)
2026-06-17

Repository: <https://github.com/sizhet/Structural-Cognitive-Emergence-SCE>

Introduction

Modern artificial intelligence has achieved remarkable progress.

AI systems can:

- write software,
- solve mathematical problems,
- generate images,
- summarize books,
- answer questions.

The progress is undeniable.

Yet a fundamental question remains:

Can an AI system become truly autonomous if it only answers questions?

This question was highlighted by Transformer co-author Lukasz Kaiser in discussions about the future of AI.

Current systems have become increasingly powerful.

However, meaningful autonomous exploration remains difficult.

Why?

Structural Cognitive Emergence proposes a simple hypothesis:

Autonomous intelligence may depend more on question generation than answer generation.

To understand why, we must first examine the role of questions in biological cognition.

The Human Learning Loop

Human cognition rarely operates as:

Question

↓

Answer

Instead, human cognition often operates as:

Observation

↓

Gap Detection

↓

Question

↓

Exploration

↓

New Structure

↓

New Gap

The system continuously generates its own objectives.

Questions are not external inputs.

Questions emerge internally.

This distinction is critical.

Why Children Appear Autonomous

A child exploring the world does not wait for assignments.

The child notices:

Cat

Dog

Bird

Moon

Rain

and begins asking:

Why?

Why not?

What is that?

How does it work?

The child's learning process is self-propelled.

Questions continuously generate new exploration paths.

This creates a natural autonomous learning loop.

Current AI Is Primarily Reactive

Most current AI systems operate differently.

The dominant pattern is:

Prompt

↓

Response

or:

Input

↓

Output

The system waits.

A question arrives.

The system answers.

This architecture is extremely useful.

However, it remains fundamentally reactive.

The system does not necessarily decide what deserves investigation.

The Missing Layer

SCE suggests that a critical layer is missing:

Gap Detection

Current AI systems often possess:

Knowledge

and

Response Generation

but lack persistent mechanisms for:

Structure Growth

Boundary Formation

Gap Discovery

Question Emergence

Without these components, autonomous inquiry remains difficult.

Questions as Internal Objectives

One of the most important properties of a question is that it creates a goal.

For example:

Why do birds fly?

immediately creates an exploration target.

The question becomes a temporary objective function.

Thus:

Question

↓

Objective

↓

Exploration

A system capable of generating its own questions may also become capable of generating its own objectives.

This is a major step toward autonomy.

Why Prompting Is Not Enough

One possible response is:

We can simply prompt AI to generate questions.

This approach is useful.

However, generated questions are not necessarily emergent questions.

There is a crucial difference.

For example:

Generate ten questions about cats.

produces questions syntactically.

But these questions may not arise from genuine structural gaps.

They are externally requested.

SCE distinguishes:

Prompted Questions

from:

Emergent Questions

The latter originate inside the cognitive structure itself.

Structural Curiosity

Many existing approaches attempt to model curiosity through:

- novelty,
- uncertainty,
- random exploration,
- reward shaping.

These approaches often produce useful behaviors.

However, SCE proposes a more structural interpretation.

Curiosity emerges when:

Graph Growth

reveals:

Unresolved Gap

The system is not exploring randomly.

The system is attempting to resolve a structural inconsistency.

Curiosity therefore becomes:

Gap-Driven Exploration

rather than:

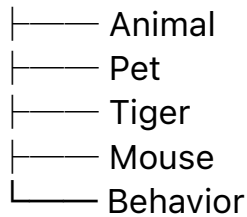
Random Exploration

Calling Graph Expansion

Suppose an AI system maintains a persistent calling graph.

For example:

CAT



As the graph grows, the system may discover:

Known Node

+

Missing Relation

For example:

Cat → Meow

Dog → Bark

Tiger → ?

The graph itself reveals incompleteness.

A question emerges:

What vocalization does a tiger produce?

The system now possesses a self-generated objective.

Differential Structures and Autonomous Inquiry

Differential structures provide another source of autonomy.

Suppose the system observes:

Bird → Fly

Bat → Fly

Cat → No Fly

The differential pattern produces tension.

The system notices:

Some mammals fly.

Most mammals do not.

A gap appears.

The system asks:

Why?

Exploration begins.

The objective emerged from structure.

Not from a user prompt.

The Gap-Question-Exploration Loop

SCE proposes that autonomous intelligence may require the following loop:

Structure Growth

↓

Gap Detection

↓

Question Emergence

↓

Exploration

↓

New Structure

The loop then repeats.

This cycle continuously generates new objectives.

The system becomes increasingly self-directed.

Why Answer Generation Alone Is Insufficient

Answer generation is essential.

But answers primarily resolve existing objectives.

Questions create new objectives.

Therefore:

Answer Generation

supports execution.

Whereas:

Question Generation

supports direction.

A system capable only of answering remains dependent on external goals.

A system capable of generating meaningful questions begins to participate in defining its own goals.

Autonomous Research

One of the most interesting future applications is autonomous scientific research.

Scientific discovery often follows:

Observation

↓

Anomaly

↓

Question

↓

Investigation

↓

Theory

The key step is not the answer.

The key step is the emergence of the question.

Future research systems may therefore require mechanisms that continuously detect:

Anomalies

Missing Connections

Structural Gaps

and convert them into research objectives.

From Question AI to Discovery AI

Current systems are largely:

Answer AI

Future systems may become:

Question AI

Eventually evolving into:

Discovery AI

The transition may proceed as:

Answer Generation

↓

Question Emergence

↓

Exploration

↓

Discovery

This progression mirrors the historical development of human cognition itself.

Human-AI Collaboration

An important implication follows.

The future may not involve replacing human curiosity.

Instead, AI may amplify it.

Humans generate questions.

AI generates additional questions.

Both systems explore together.

The result becomes:

Human

+

AI

=

Expanded Question Space

The partnership becomes a collaborative engine of discovery.

Revisiting the Child

The journey began with a simple observation:

A child sees a few cats.

A concept emerges.

Soon afterward:

Why?

appears.

This tiny question may contain the seeds of autonomy.

The child is no longer merely observing.

The child is directing exploration.

The child is generating objectives.

The child is becoming an autonomous learner.

The Central Hypothesis

The central hypothesis of this document is:

Meaningful autonomous intelligence requires the emergence of internally generated questions.

In SCE terms:

Structure

↓

Gap

↓

Question

↓

Exploration

↓

Structure

forms a self-sustaining growth loop.

Answer generation alone is not sufficient.

Autonomy begins when a system starts generating its own questions.

Next

SCE-010

Structural Cognitive Emergence

A Unified Framework for Concept Formation, Question Emergence, Cognitive Growth, and Autonomous Intelligence.